CLINICAL PHARMACOKINETICS: PRINCIPLES

Logarithm of 2

Volume of distribution

Clearance

$$t½ = \frac{0.693 \times \mathrm{Vd}}{CL}$$

Half-life

BPM1 PHARMACOLOGY

Fall Term 2025

St. George's University
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OBJECTIVES

SOM.MK.1.BPM1.1.FTM.4.PHAR.0026 Describe the aim of clinical pharmacokinetics.

SOM.MK.1.BPM1.1.FTM.4.PHAR.0027 Define volume of distribution. Describe the factors that affect the volume of distribution of a drug.

SOM.MK.1.BPM1.1.FTM.4.PHAR.0028 Explain the uses of the volume of distribution. Determine the volume of distribution of a drug. Recognize the distribution phase and the elimination phase in a plot of plasma drug concentration following an IV bolus.

SOM.MK.1.BPM1.1.FTM.4.PHAR.0029 Define clearance. Distinguish between clearance and rate of drug elimination

SOM.MK.1.BPM1.1.FTM.4.PHAR.0030 Define first-order kinetics of drug elimination. Explain why most drugs follow first-order kinetics of elimination. Explain why clearance is constant for drugs that follow first-order kinetics of elimination.

SOM.MK.1.BPM1.1.FTM.4.PHAR.0031 Define half-life. Calculate the drug plasma levels reached after a given number of half-lives during drug administration, and after drug administration is discontinued. Calculate the half-life of a drug graphically from a plot of the drug plasma level vs time. Describe the relationship between half-life, volume of distribution, and clearance.

SOM.MK.1.BPM1.1.FTM.4.PHAR.0032 Describe the factors that affect half-life

SOM.MK.1.BPM1.1.FTM.4.PHAR.0033 Define zero-order kinetics of drug elimination. Explain why some drugs follow zero-order kinetics of elimination. List examples of drugs that follow zero order kinetics of elimination

CLINICAL PHARMACOKINETICS

- Clinical pharmacokinetics aims to design dosage regimens which:
  - Optimize therapeutic response of a drug
  - Minimize chance of adverse reactions

CLINICAL PHARMACOKINETICS

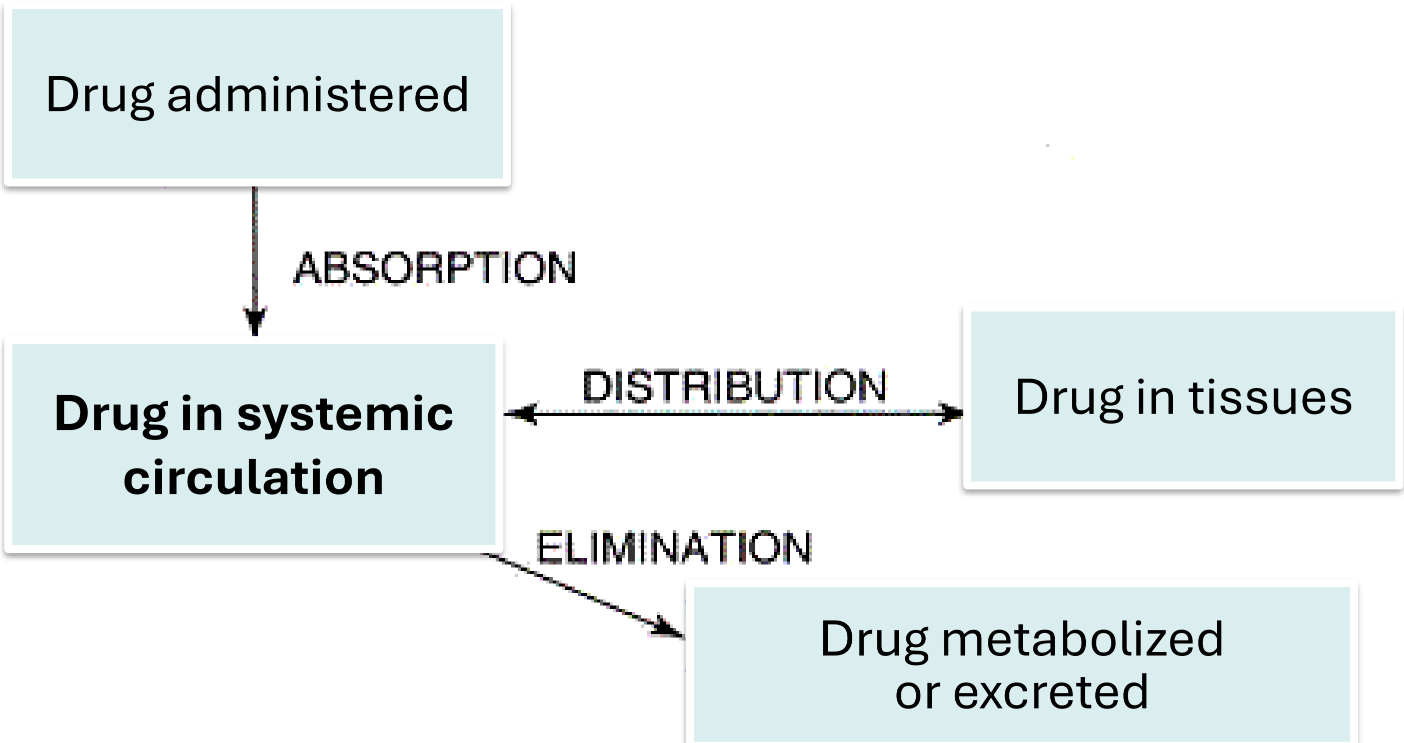
Fundamental tenet:

A relationship exists between the effects of a drug and the concentration of the drug in the blood.

CLINICAL PHARMACOKINETICS

- The plasma concentration of a drug is a function of
    - The rate of input of the drug into the plasma
    - The rate of distribution of the drug to the tissues
    - The rate of elimination of the drug

CLINICAL PHARMACOKINETICS



PHARMACOKINETIC PARAMETERS

- The three most important pharmacokinetic parameters are:
    - **Volume of distribution**: a measure of the apparent space in the body available to contain the drug
    - **Clearance**: a measure of the body's ability to eliminate the drug
    - **Bioavailability:** the fraction of drug absorbed into the systemic circulation.

VOLUME OF DISTRIBUTION (Vd)

- Relates the amount of drug in the body to the concentration of drug in the plasma:

$$\mathbf{Vd} = \frac{\textbf{Amount of drug in the body}}{\textbf{Plasma drug concentration}}$$

- Units = volume

VOLUME OF DISTRIBUTION (Vd)

- The Vd doesn’t necessarily refer to an identifiable physiological volume.

VOLUME OF DISTRIBUTION (Vd)

- **Vd is the volume that would be required to contain all of the drug that is in the body at the same concentration as it is in the blood.**

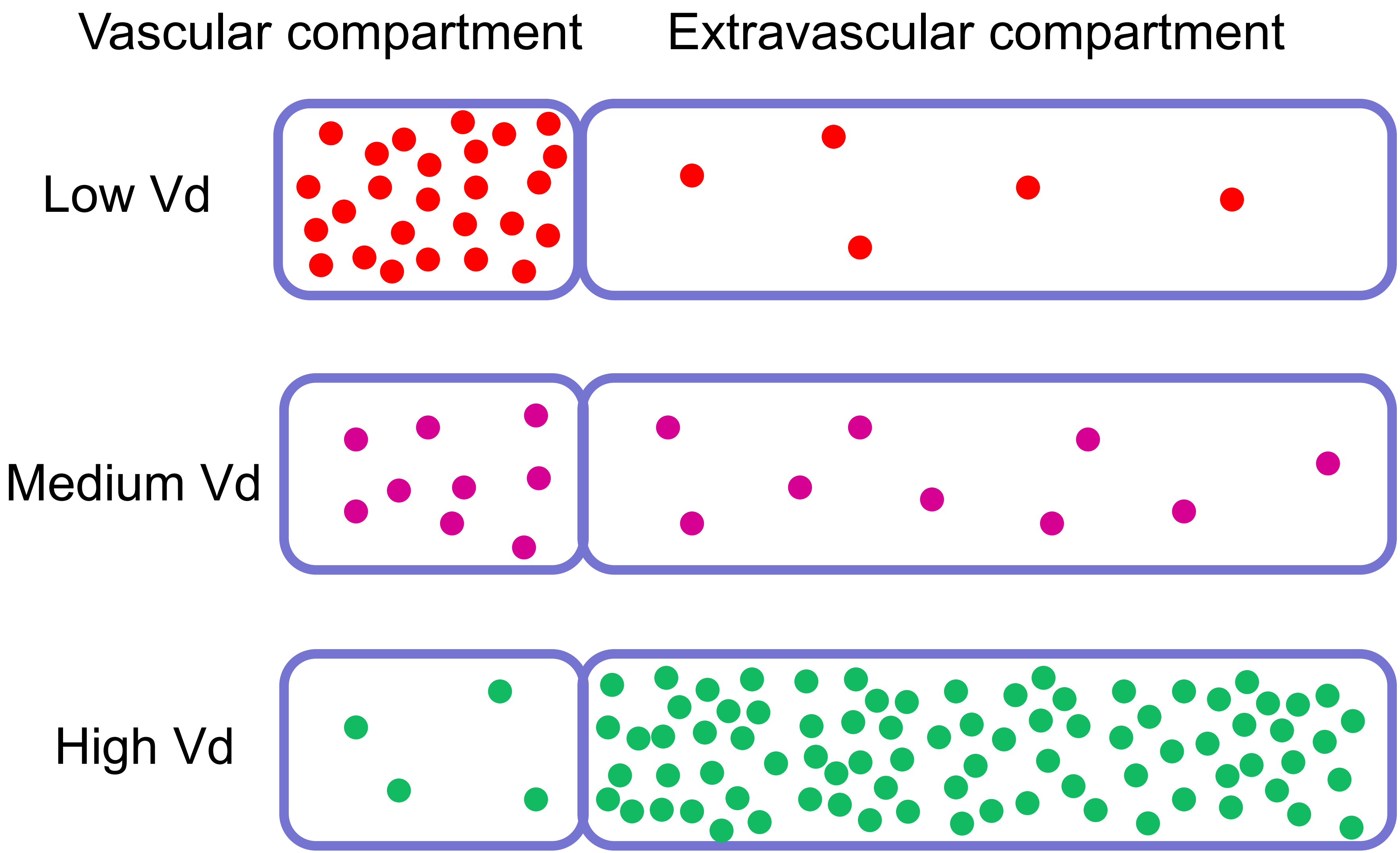
VOLUME OF DISTRIBUTION (Vd)

- Drugs that are completely retained within the vascular compartment have a minimum Vd equal to the vascular compartment.
- Drugs that have much higher concentrations in the extravascular tissue than in the vascular compartment have a very high Vd.

VOLUME OF DISTRIBUTION (Vd)

- If a drug is bound to peripheral tissues, or partitioned into body fat its concentration in plasma may be very low.
- As a result, the Vd may be very high.
- For example, the Vd for quinacrine is **50,000 L** in a person whose physical body volume is 70L.

VOLUME OF DISTRIBUTION (Vd)



VOLUME OF DISTRIBUTION (Vd)

- The Vd is a **proportionality factor** that relates the plasma concentration of drug to the amount of drug in the body.
- **The volume of distribution allows us to convert concentrations to amounts.**

VOLUME OF DISTRIBUTION (Vd)

- **The main use of the volume of distribution is to determine the loading dose to quickly achieve a target plasma concentration.**

DETERMINATION OF Vd

- To calculate Vd, we need to know the amount of drug in the body and the plasma drug concentration.

DETERMINATION OF Vd

- The only time the amount of drug in the body is known accurately is immediately after the drug has been given IV, because this is the dose.

DETERMINATION OF Vd

- If we can determine the plasma concentration at time 0 (C_0), then we can calculate the Vd as follows:

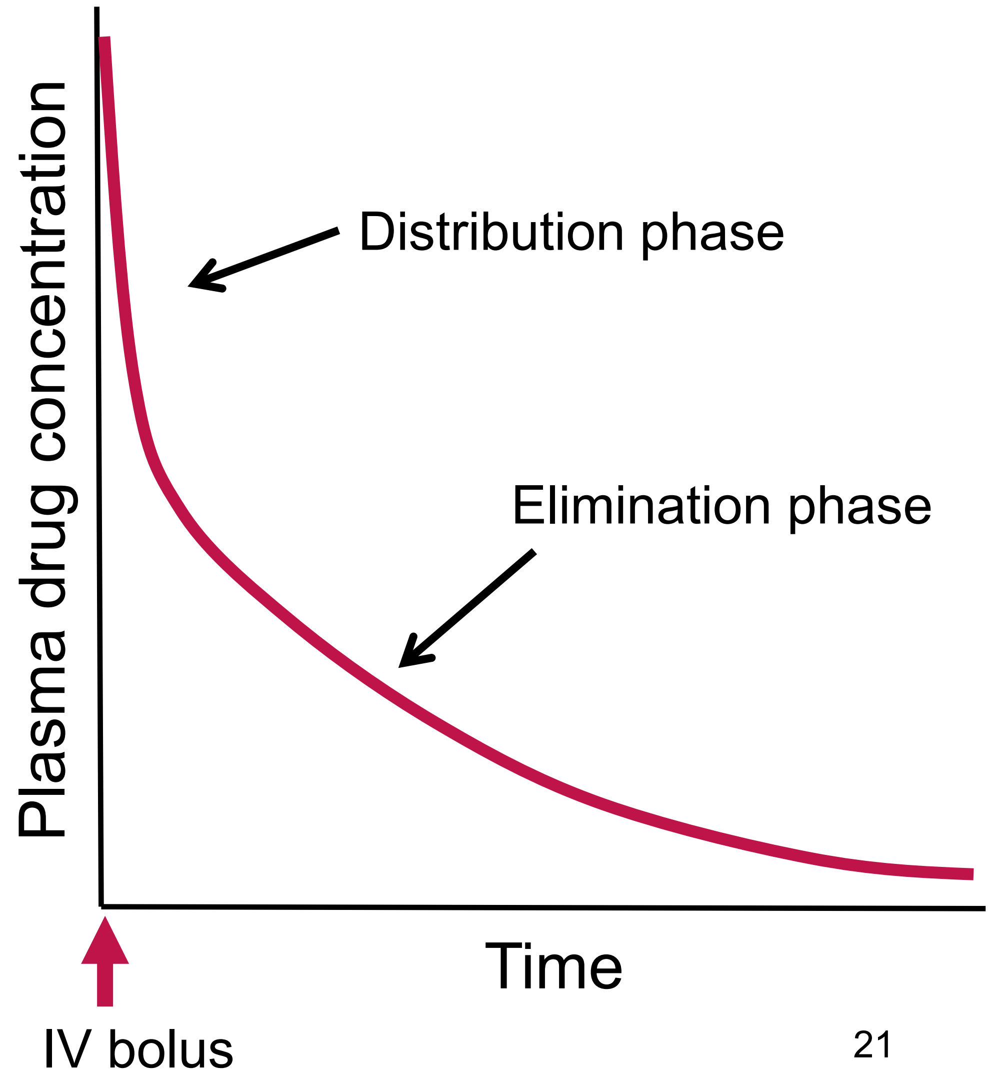
$$V_d = \frac{\text{Dose}}{C_0}$$

DETERMINATION OF Vd

- A dose of a drug is injected IV.
- The plasma drug concentration is plotted vs time.
- Usually, the plot shows two phases.

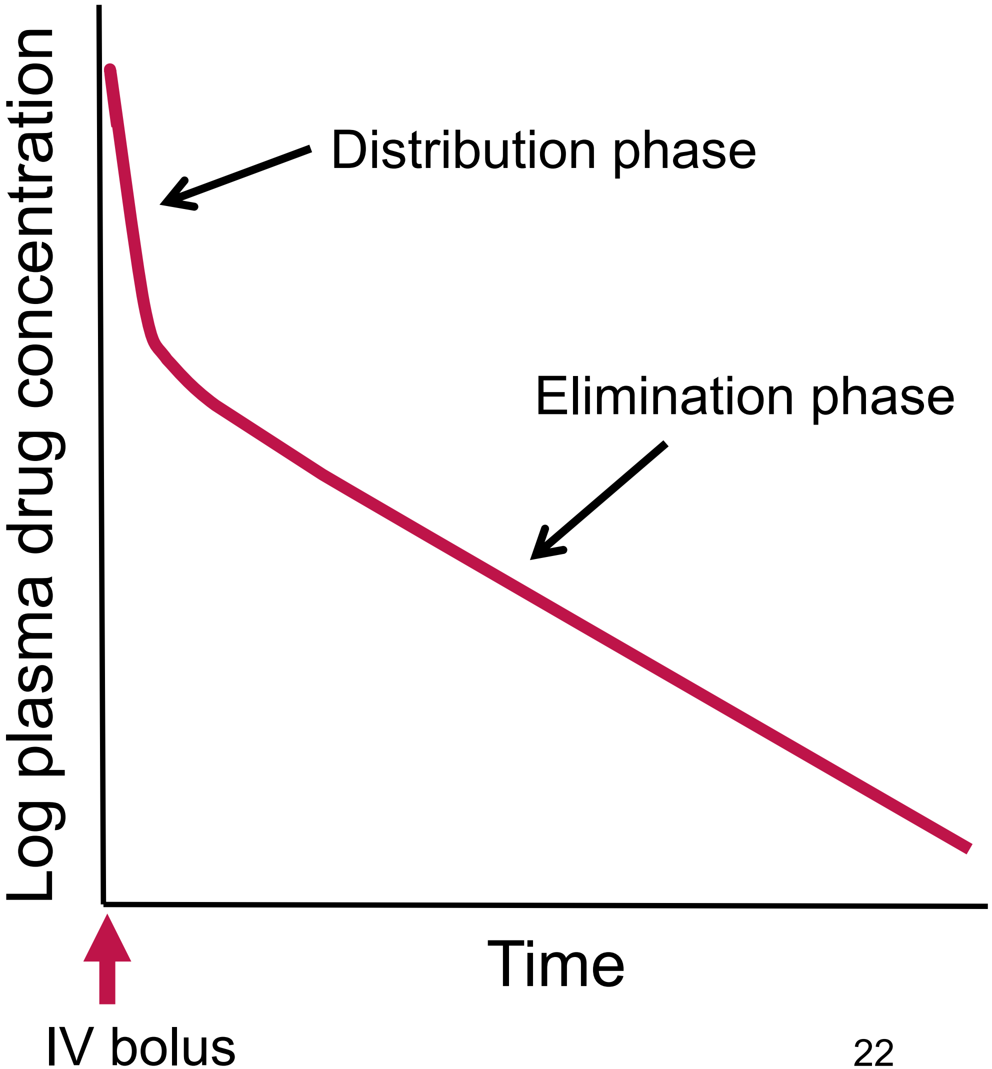
DETERMINATION OF Vd

- The rapid fall is the distribution phase.
- The slower phase is the elimination phase.



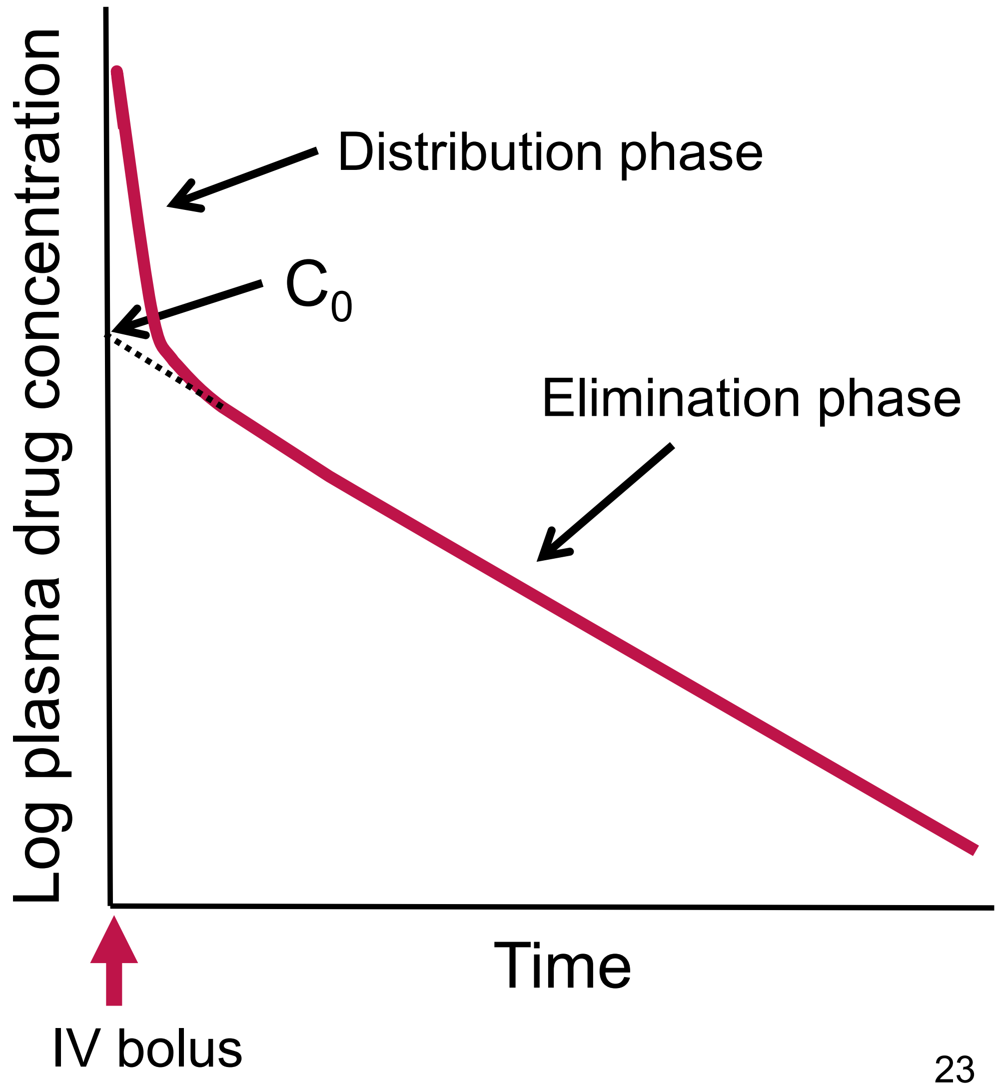
DETERMINATION OF Vd

- We linearize the curve by plotting the logarithm of the plasma drug concentration vs time.



DETERMINATION OF Vd

Extrapolation of the elimination curve to the y axis, yields the concentration that would have existed at the start if the dose had been instantly distributed.



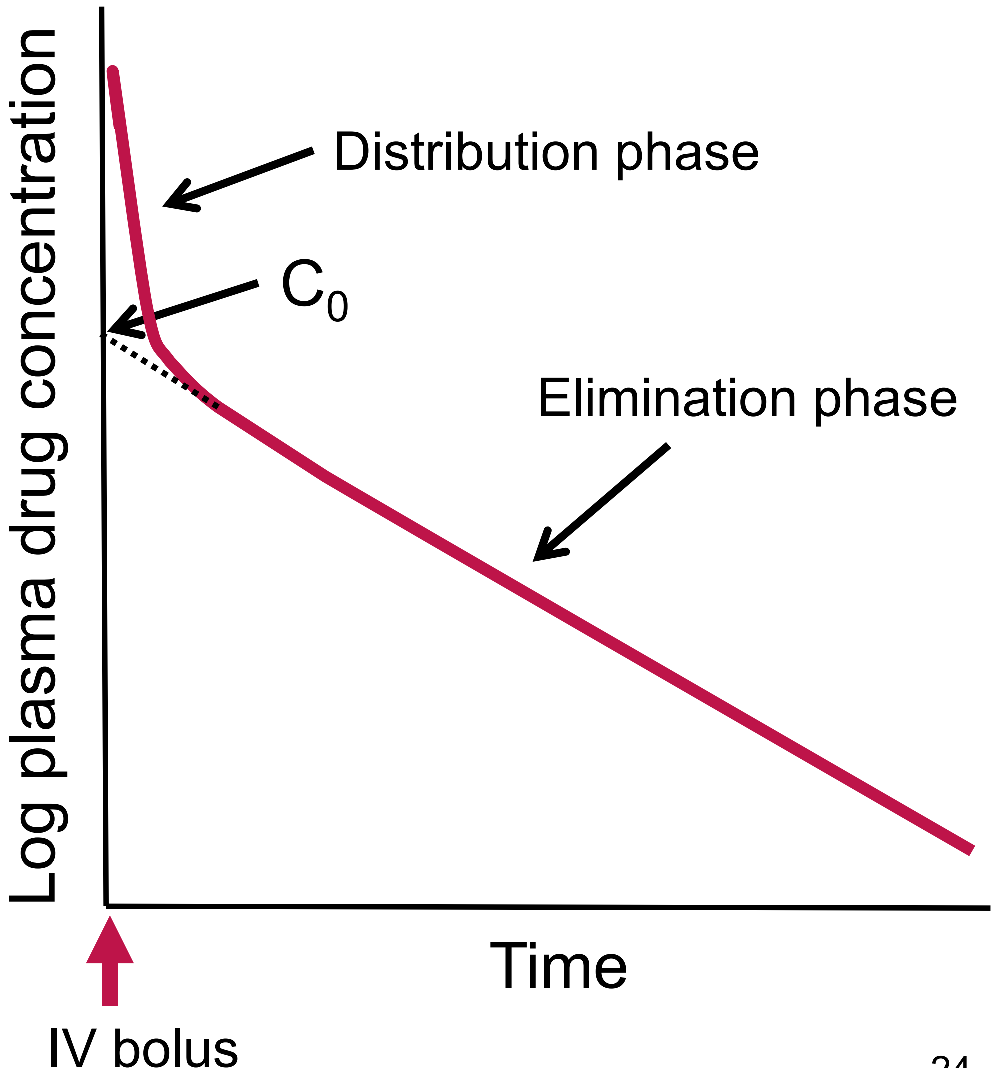
DETERMINATION OF Vd

- The initial concentration is:

$$C_0 = \frac{\textbf{Dose}}{V_d}$$

So,

$$V_d = \frac{\textbf{Dose}}{C_0}$$



DETERMINATION OF Vd

- So, if we want to achieve quickly a target plasma drug concentration (TC), we can calculate the dose required as follows:

Dose = Vd X TC

CLEARANCE (CL)

- Clearance is defined as the volume of blood cleared of drug per unit time.
- Clearance predicts the rate of elimination of the drug in relation to the drug concentration:

$$\mathbf{CL} = \frac{\textbf{Rate of elimination of drug (amount/time)}}{\textbf{Plasma drug concentration (amount/volume)}}$$

- Units = volume per unit time

CLEARANCE (CL)

- Total body clearance is the sum of all the different clearance processes occurring for a given drug.
- The two major sites of drug elimination are the **kidneys and the liver.**

CLEARANCE (CL)

- For most drugs, CL is **constant** over the plasma concentrations used clinically.
- Therefore, the rate of drug elimination is directly proportional to the drug concentration:

$$\boxed{\textbf{Rate of elimination} = \textbf{CL} \times \textbf{C}}$$

- **This is called “first order” elimination.**

CLEARANCE (CL)

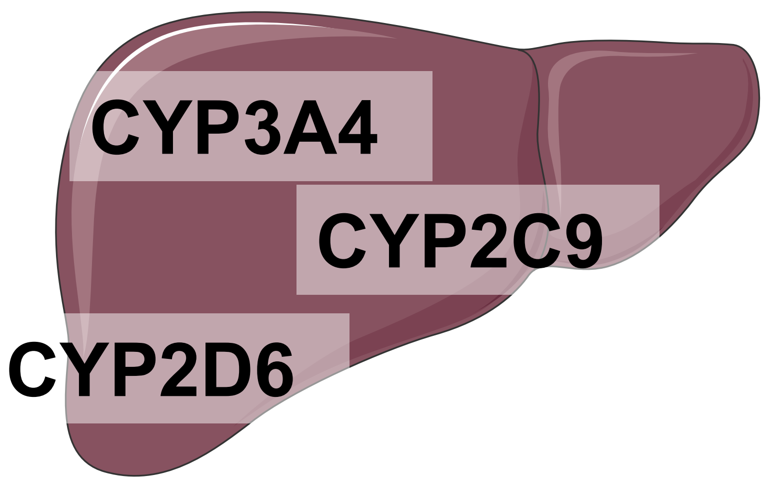
Rate of elimination = CL x C

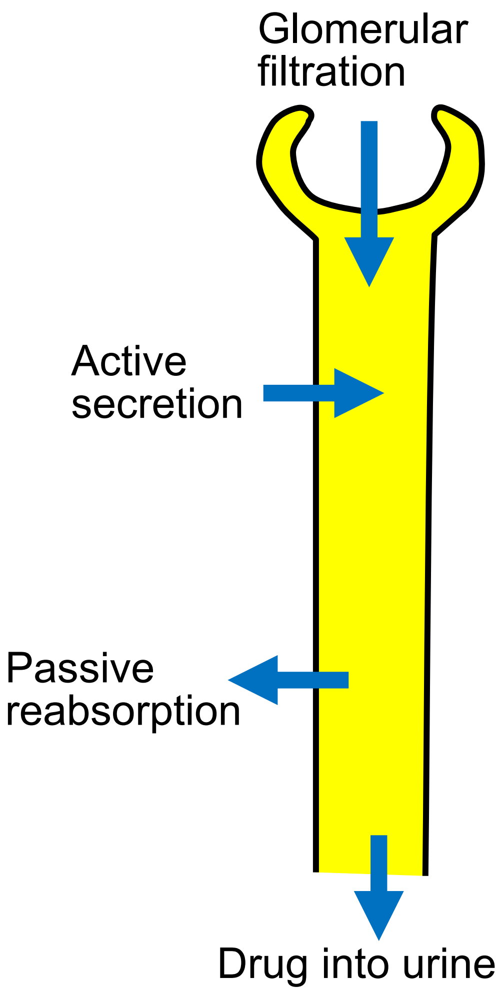
- For a drug eliminated with **first-order kinetics**, **CL is a constant**, ie, the ratio of the rate of elimination to the plasma concentration is the same regardless of plasma concentration.

$$\mathbf{CL} = \frac{\textbf{Rate of elimination}}{\mathbf{C}}$$

Why do most drugs follow first order kinetics?

- This occurs because the physiological mechanisms of drug elimination (enzymes and transporters) **are not saturated**.



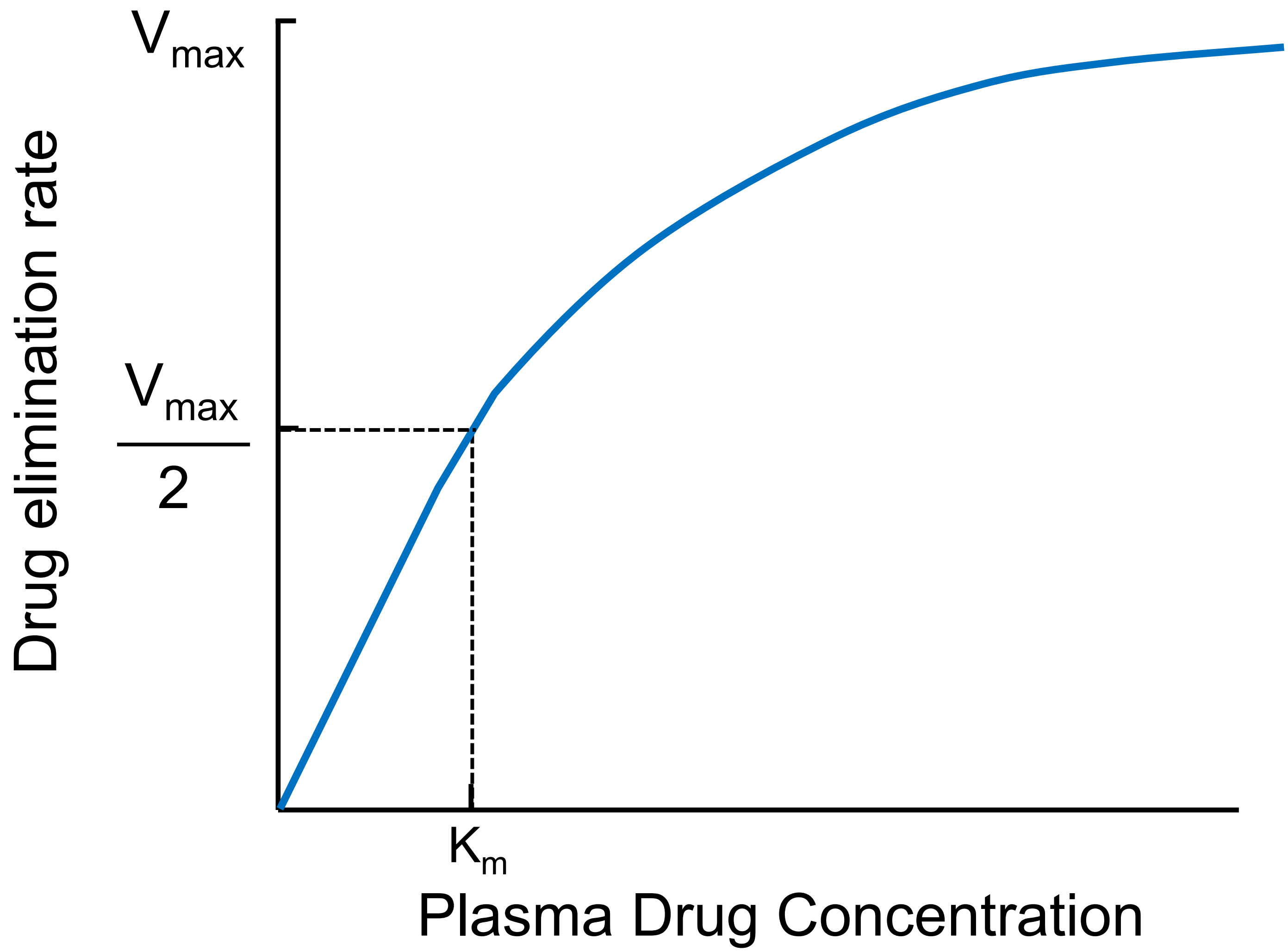


Why do most drugs follow first order kinetics?

- The dependence of the rate of drug elimination on drug plasma concentration is described by the equation:

$$\textbf{Rate of elimination} = \frac{V_{max} \times C}{K_m + C}$$

Why do most drugs follow first order kinetics?



Why do most drugs follow first order kinetics?

- When the drug concentration is very low compared to the K_m ($C \ll K_m$), the equation reduces to:

$$\textbf{Rate of elimination} = \frac{V_{max} \times C}{K_m}$$

Where $$\textbf{CL} = \frac{V_{max}}{K_m}$$

Why do most drugs follow first order kinetics?

- So, the rate of elimination becomes:

Rate of elimination = CL x C

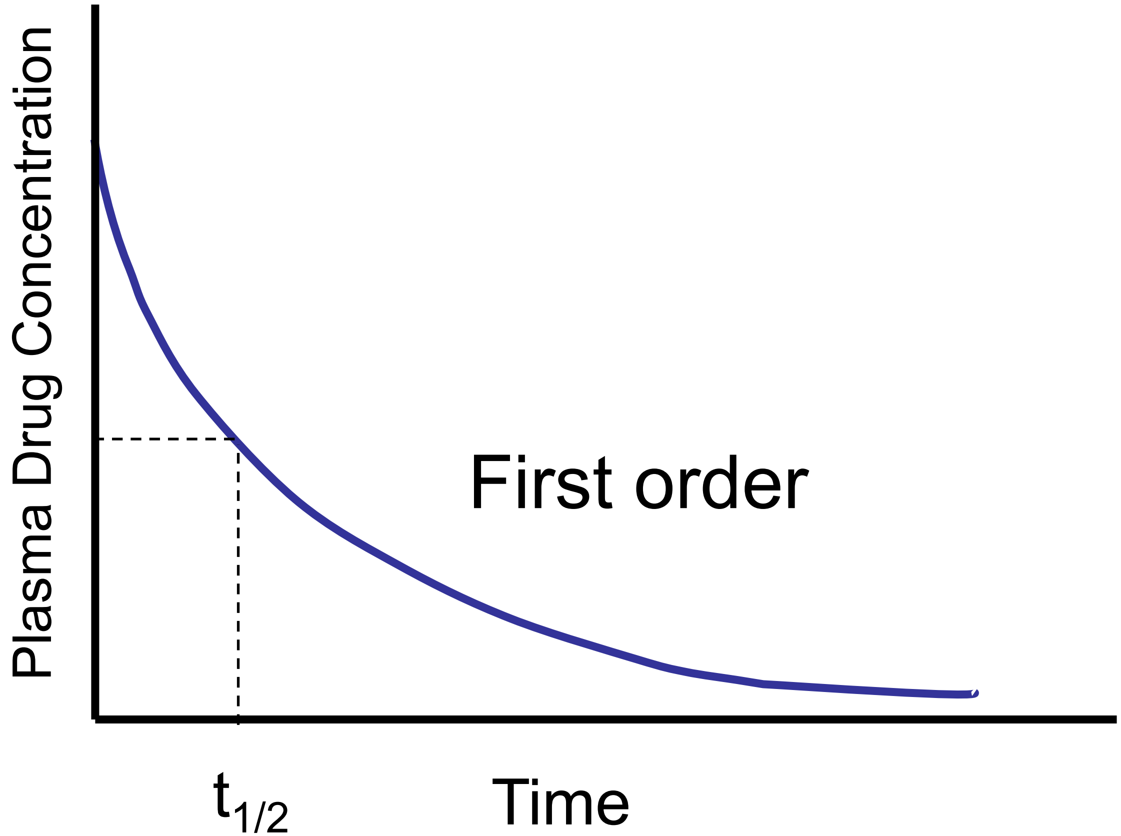
- For the wide majority of drugs the therapeutic plasma concentration is well below the K_m, therefore first order elimination is observed.

FIRST ORDER KINETICS: PROPERTIES

- The initial concentration is: $$C_0 = \frac{\text{Dose}}{V_d}$$
- The concentration at a later time will depend on the rate of elimination of the drug.

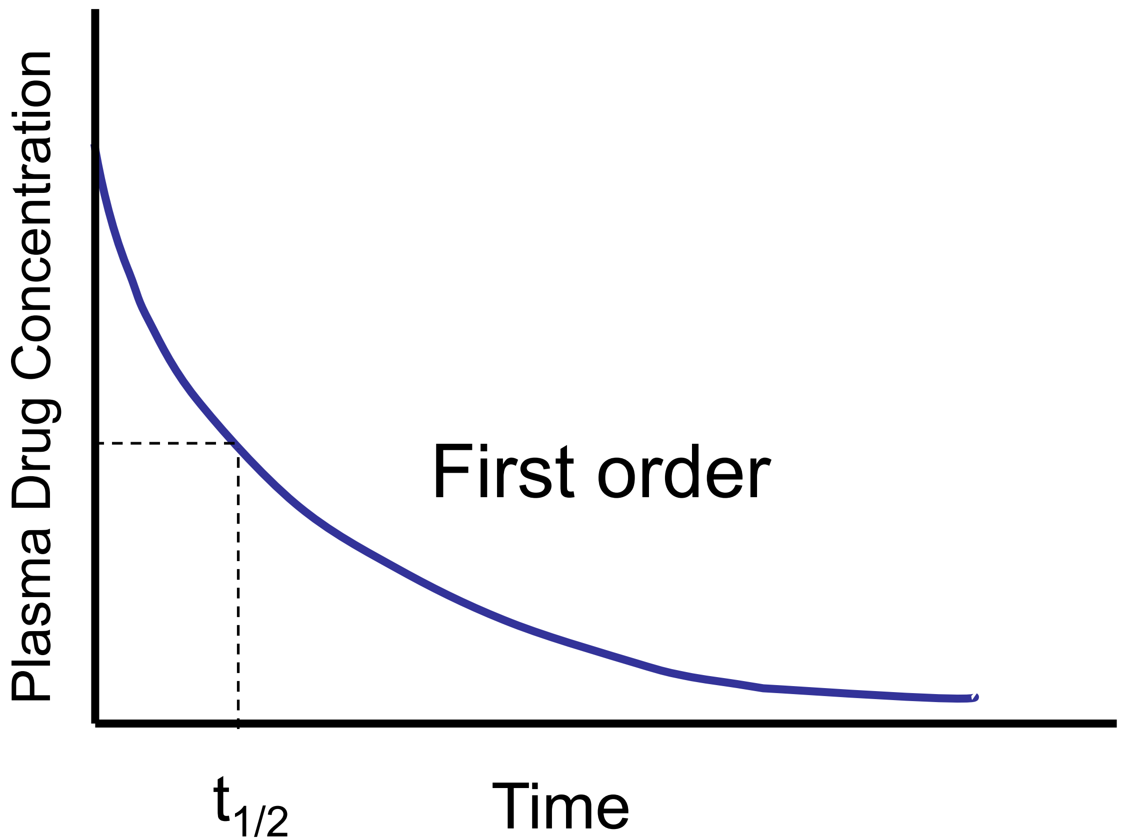
FIRST ORDER KINETICS: PROPERTIES

- The drug concentration decays exponentially.



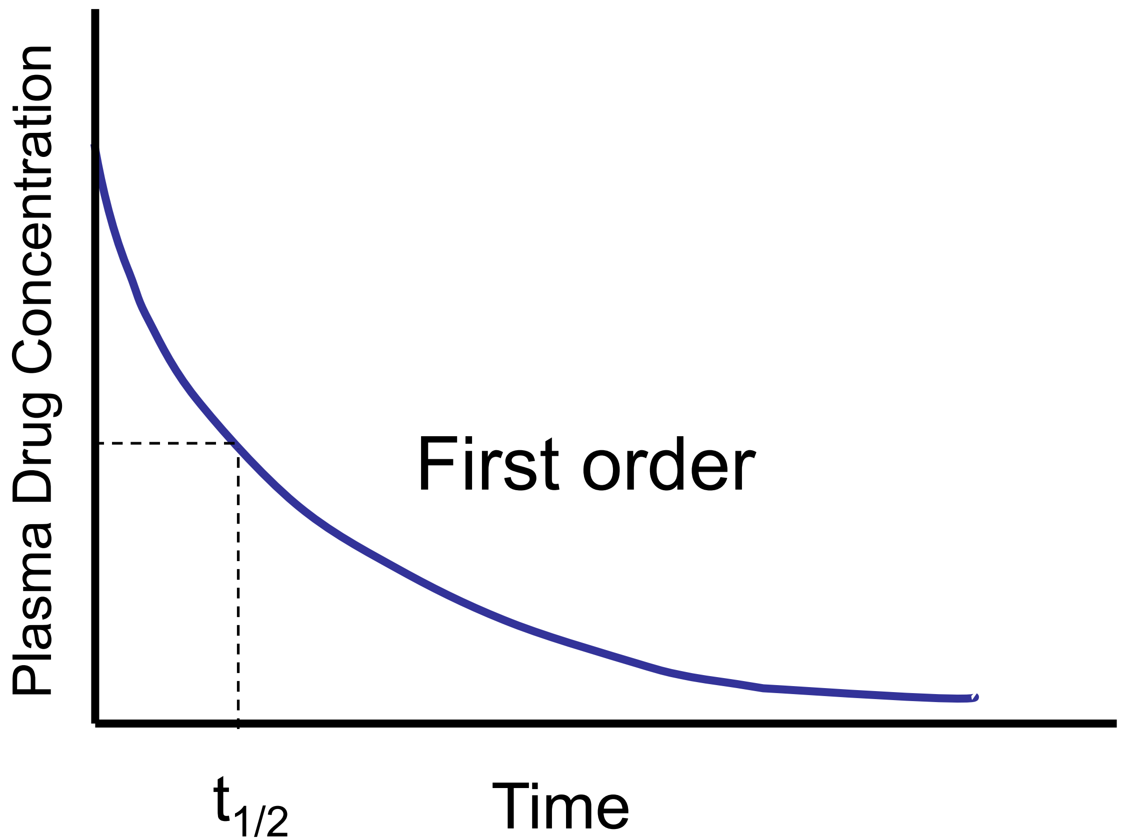
FIRST ORDER KINETICS: PROPERTIES

- For drugs with first-order kinetics of elimination the rate of elimination is directly proportional to drug concentration.



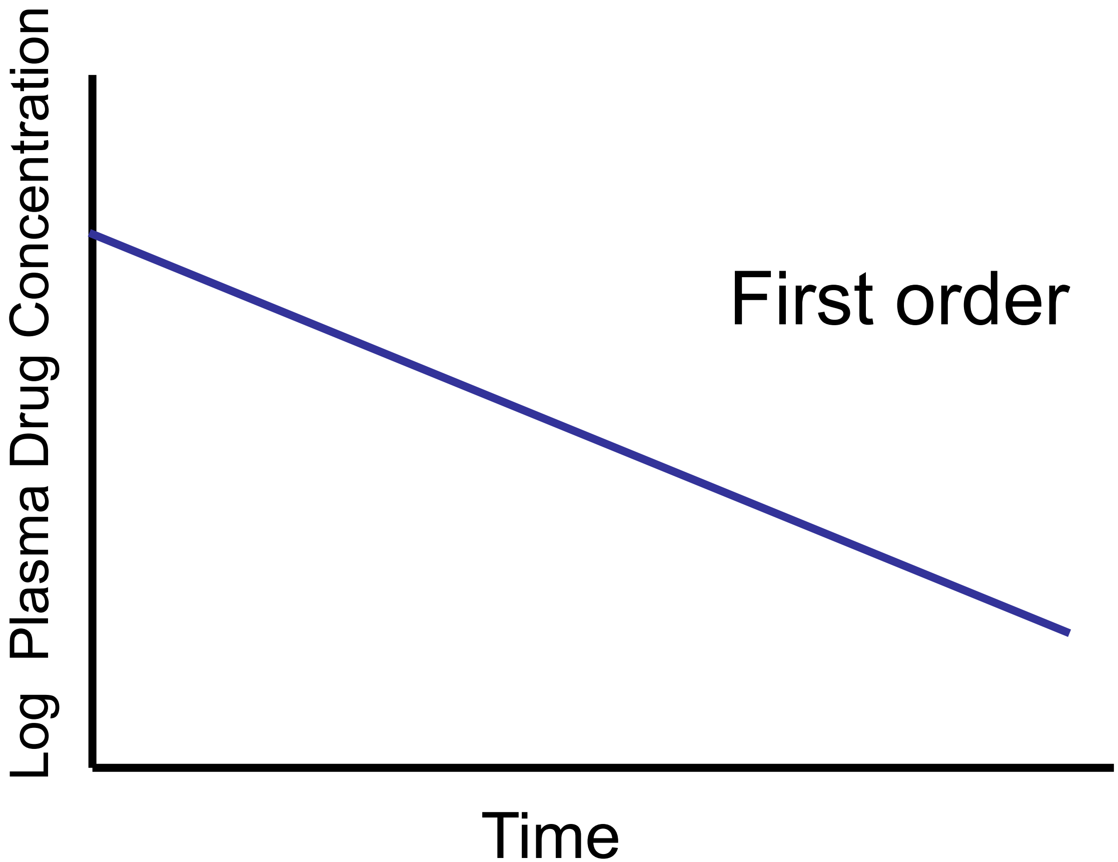
FIRST ORDER KINETICS: PROPERTIES

- When a drug follows first-order kinetics, a constant fraction of the drug is eliminated per unit of time.



FIRST ORDER KINETICS: PROPERTIES

- Plotting the logarithm of C against t yields a straight line.



HALF-LIFE

- Half-life is the time required to reduce the amount of drug in the body by 50% during elimination.
- Half-life is a constant for drugs eliminated by first-order kinetics (the great majority of drugs).

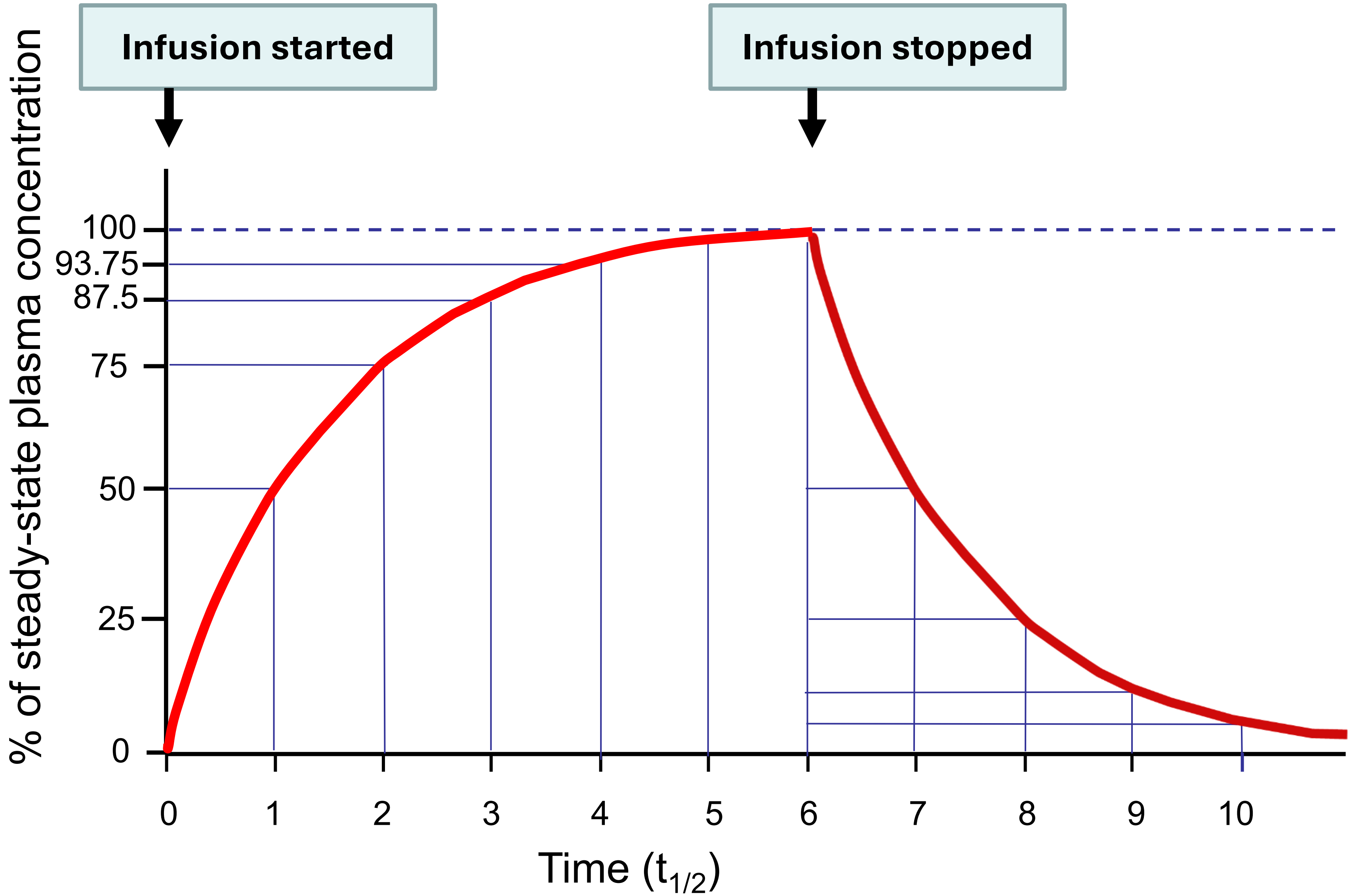
HALF-LIFE

- Half-life determines the rate at which blood concentration rises during a constant infusion and falls after infusion is stopped.

HALF-LIFE

- During constant IV infusion, every half-life the drug concentration increases 50% of the difference between the C_{ss} and the current C.
- During elimination, every half-life the drug concentration is reduced 50%.

HALF-LIFE



Infusion started
Infusion stopped
% of steady-state plasma concentration
100
93.75
87.5
75
50
25
0
0
1
2
3
4
5
6
7
8
9
10
Time ($t_{1/2}$)

HALF-LIFE

During a constant IV infusion of a drug:

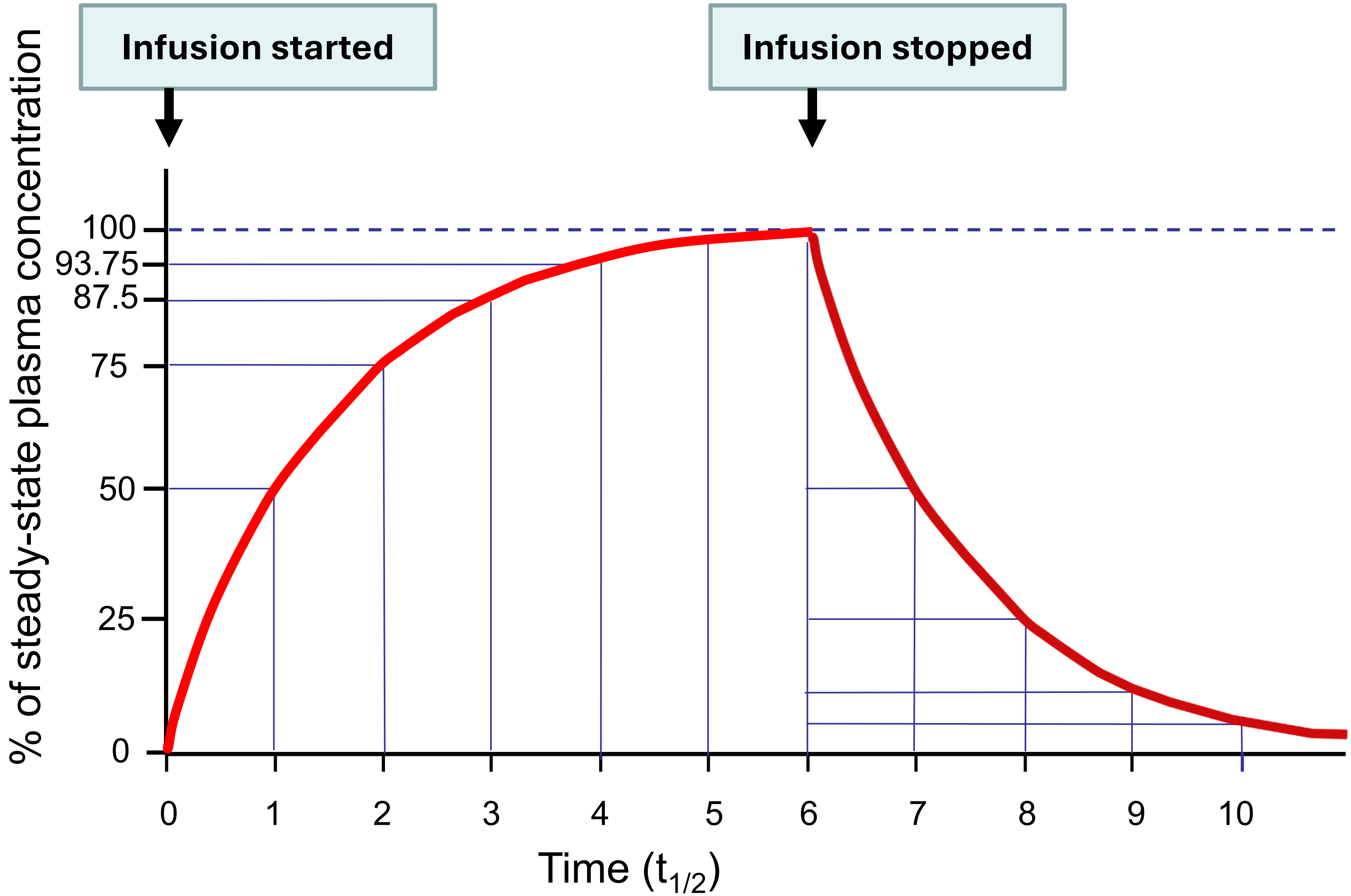
- 50% of C_{ss} is reached after 1 half-life
- 75% after 2 half-lives
- 87.5% after 3 half-lives
- 93.75% after 4 half-lives

HALF-LIFE

After stopping the IV infusion the drug:

- 50% of the drug is lost after 1 half-life
- 75% after 2 half-lives
- 87.5% after 3 half-lives
- 93.75% after 4 half-lives

HALF-LIFE



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Time ($t_{1/2}$)

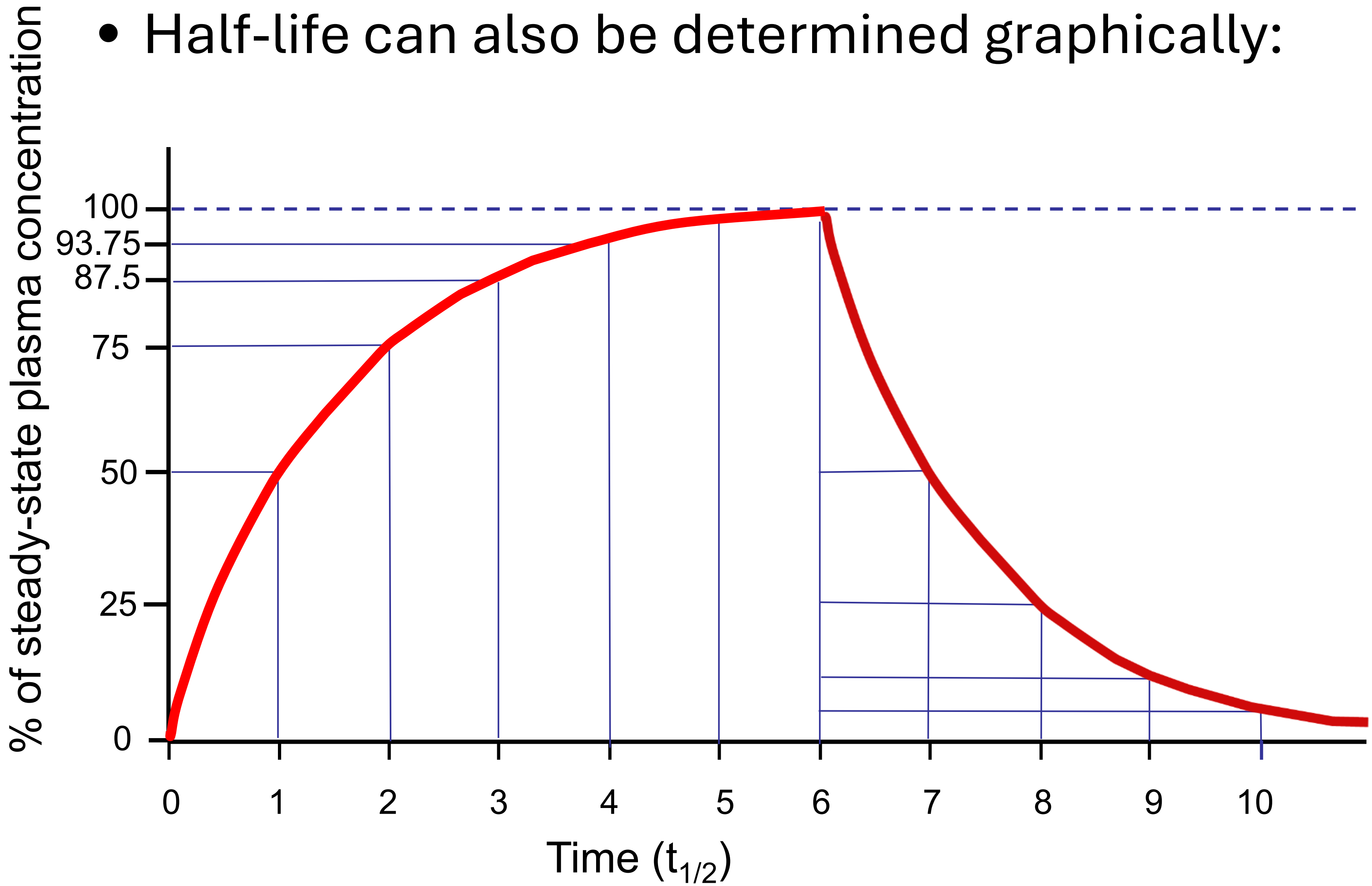
HALF-LIFE

$$t_{1/2} = \frac{0.693 \times V_d}{CL}$$

- **Units = time**

HALF-LIFE

- Half-life can also be determined graphically:

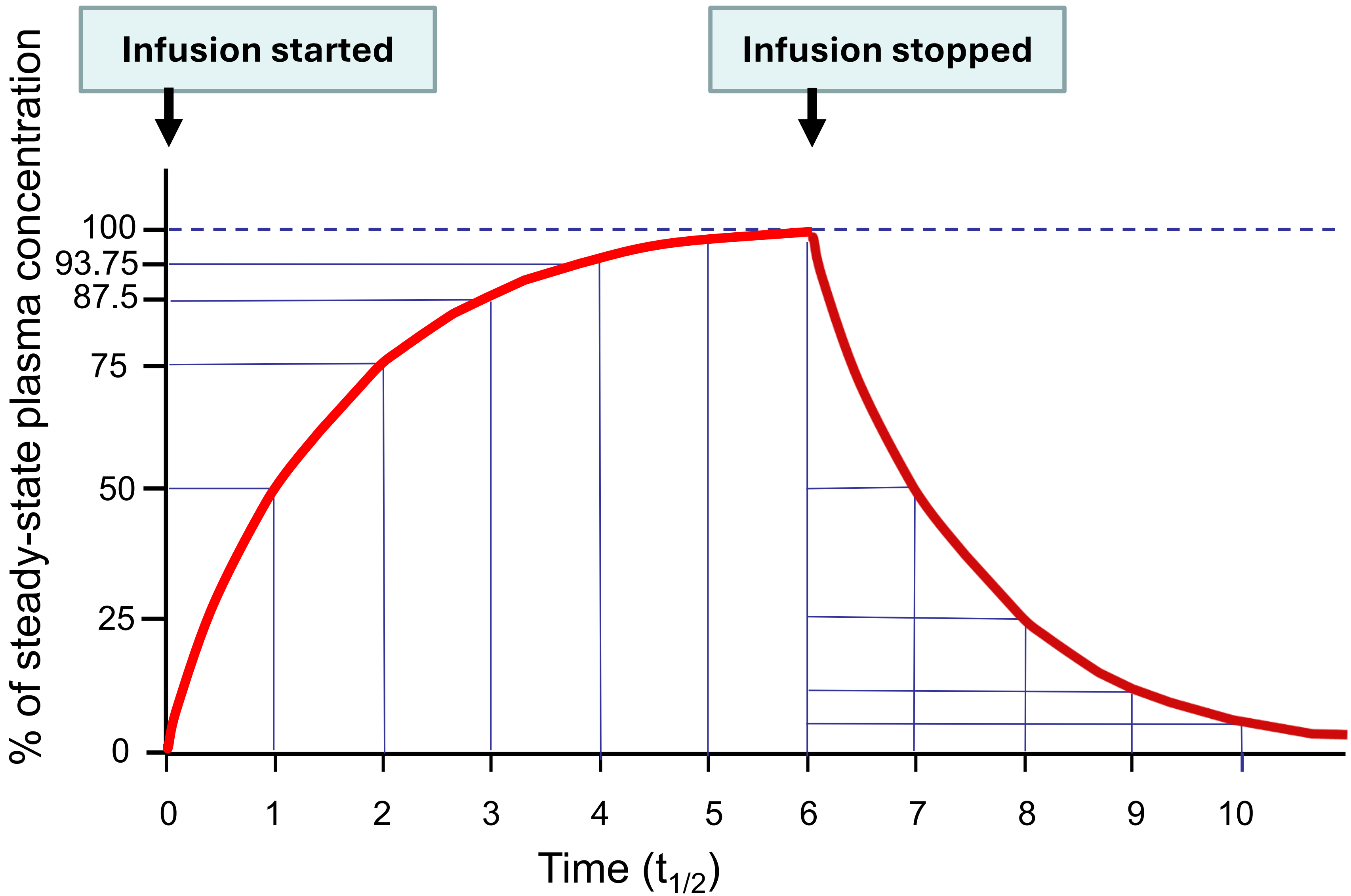


HALF-LIFE

- It can be assumed that steady state is attained after about 4 half-lives.

- It can be assumed that the drug has been effectively eliminated after about 4 half-lives.

HALF-LIFE



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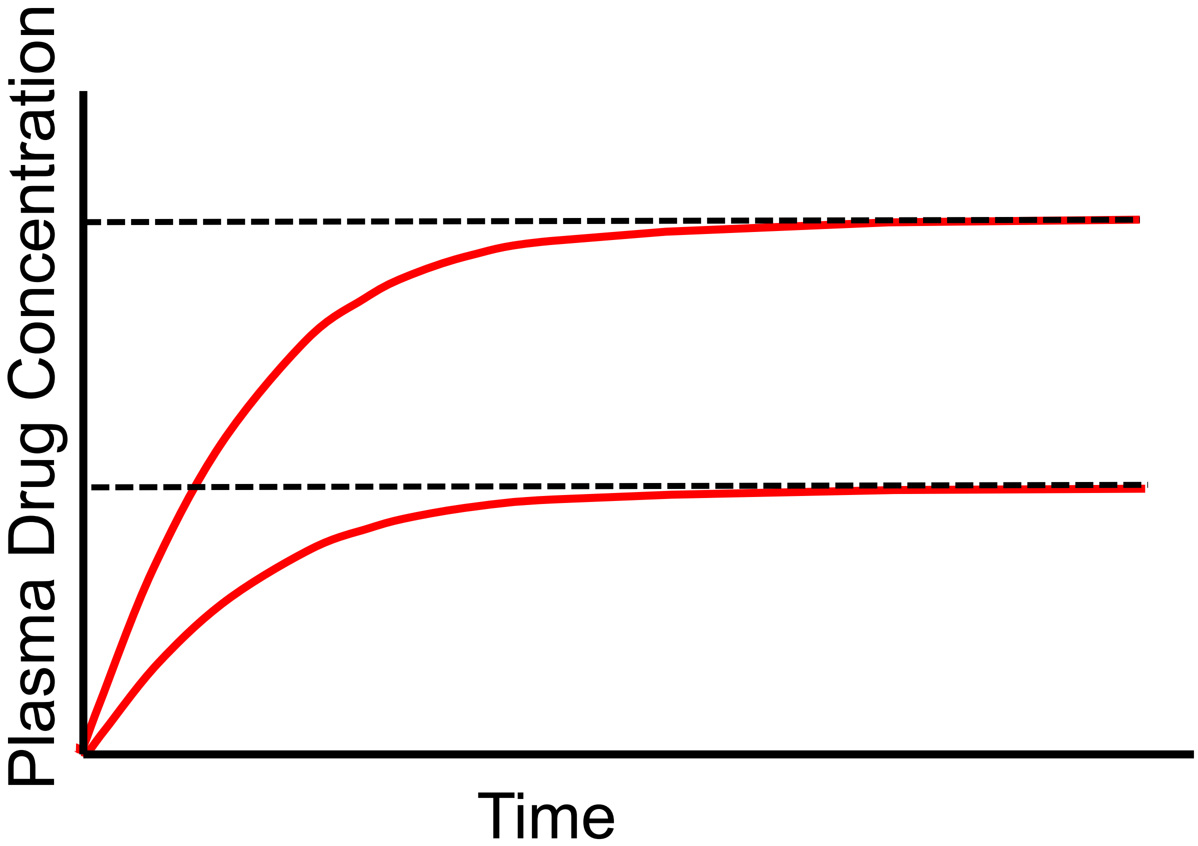
Time ($t_{1/2}$)

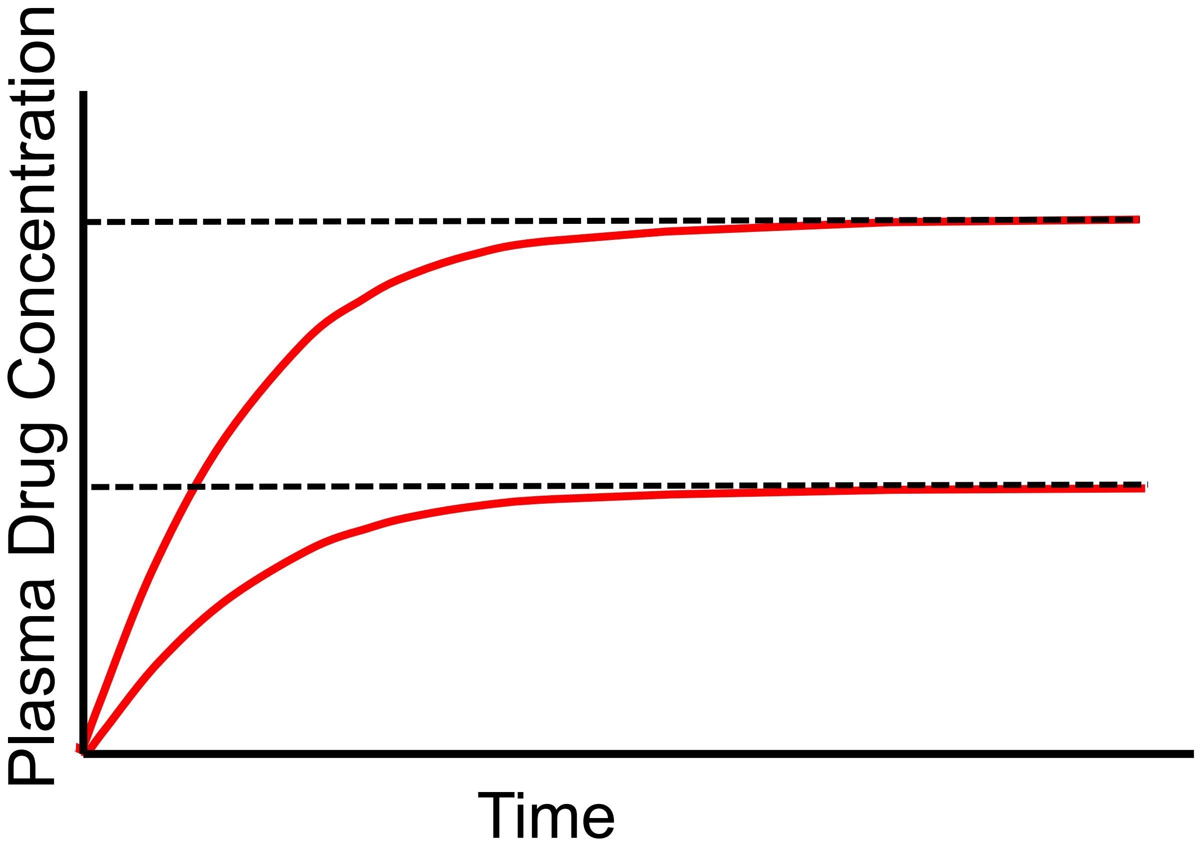
HALF-LIFE

- Doubling the infusion rate doubles the Css.



HALF-LIFE

- Doubling the infusion rate doubles the Css.
- But increasing the infusion rate does not influence the time required to reach C_{ss}.



FACTORS AFFECTING HALF-LIFE

Factor affecting half-life	Effect on half-life
Effects on Vd	
Obesity	Increased
Pathologic fluid	Increased
Effects on CL	
CYP induction	Decreased
CYP inhibition	Increased
Ageing	Increased
Cardiac failure	Increased
Liver failure	Increased
Renal failure	Increased

KINETICS OF DRUG ELIMINATION: SATURATION KINETICS

- Some drugs exhibit saturation kinetics of elimination.

Why do some drugs exhibit saturation kinetics?

- Drug metabolism and tubular secretion are saturable processes.
- When drug concentration exceeds K_m, nonlinear kinetics is observed.

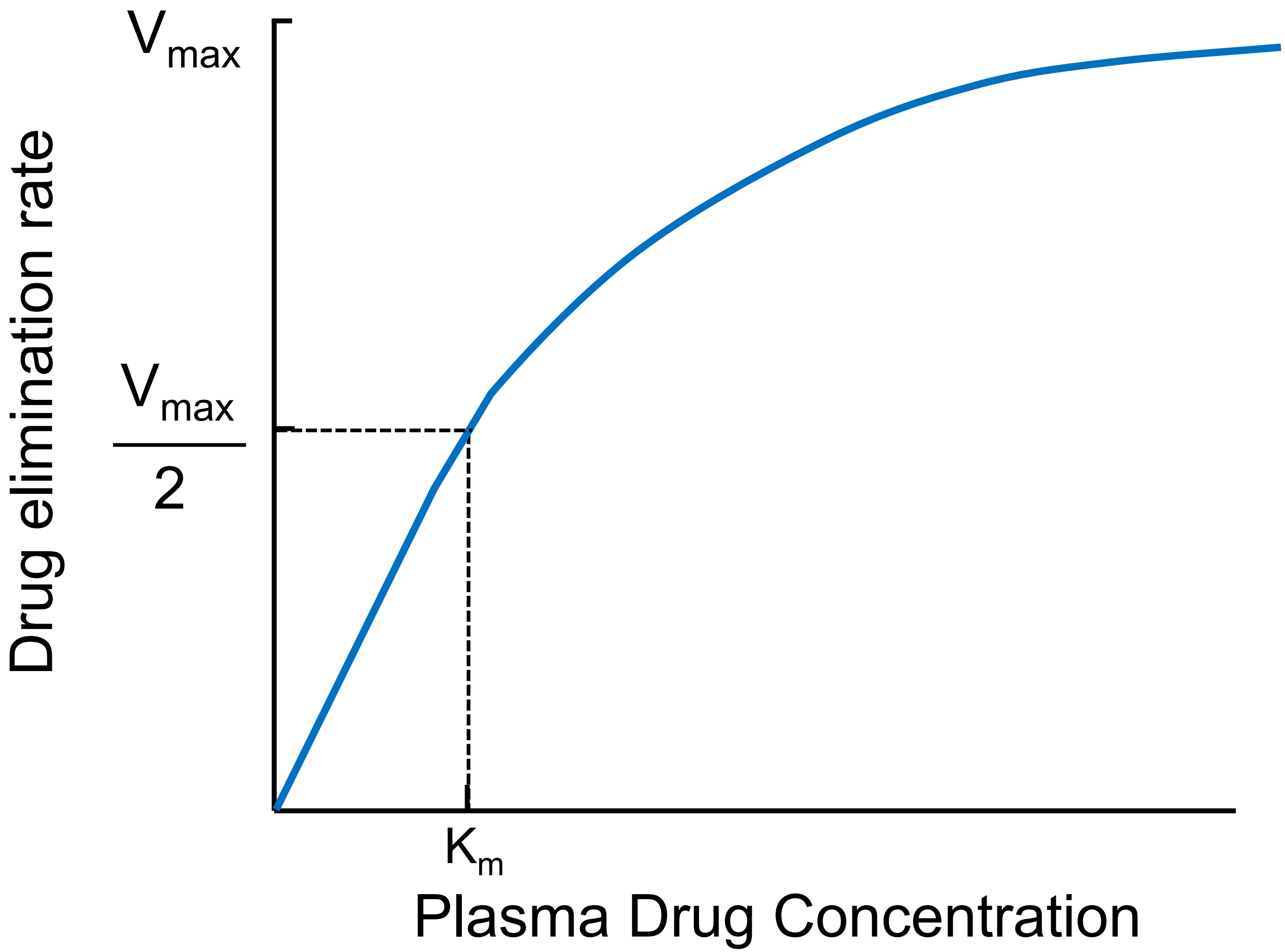
Why do some drugs exhibit saturation kinetics?

$$\textbf{Rate of elimination} = \frac{V_{max} \times C}{K_m + C}$$

- When $C >> K_m$ the equation reduces to:

$$\textbf{Rate of elimination} = \frac{V_{max} \times C}{C} = V_{max}$$

Why do some drugs exhibit saturation kinetics?



Which drugs exhibit saturation kinetics?

- Zero order elimination is observed with a small number of drugs:
    - *Aspirin at high doses*
    - *Ethanol*
    - *Phenytoin*

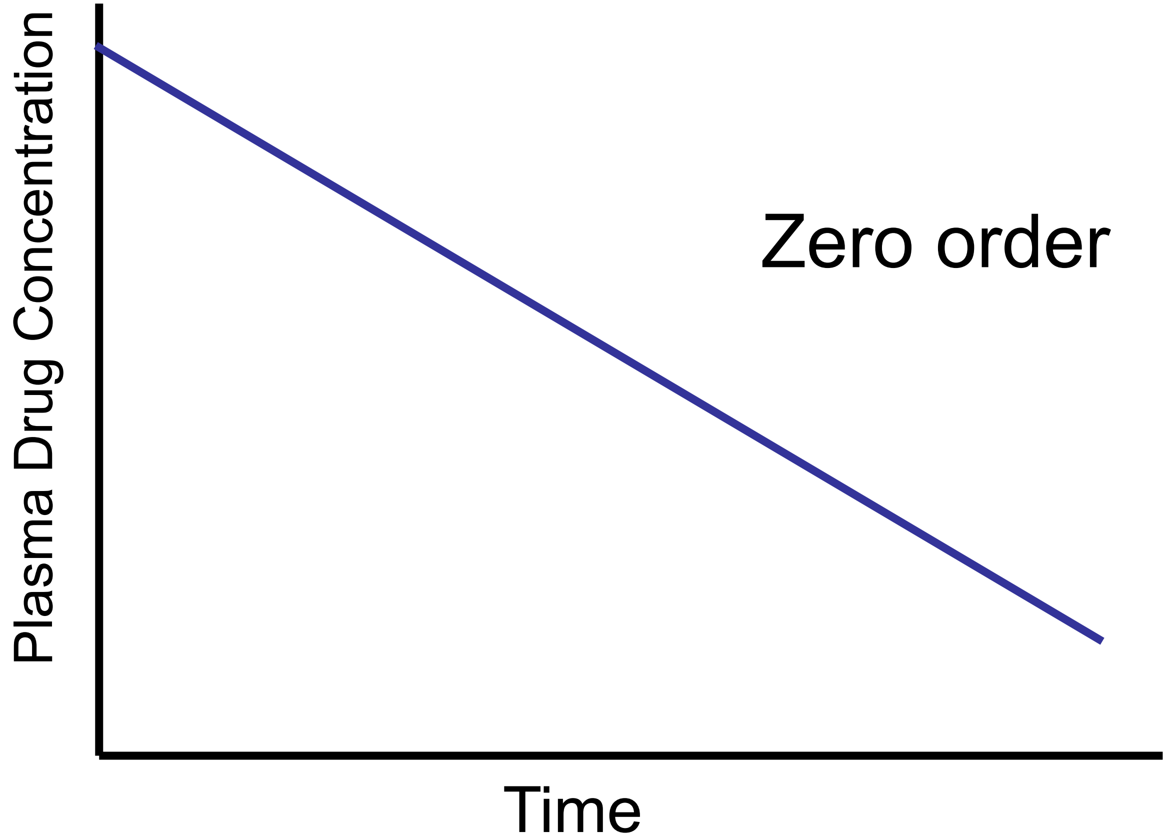
SATURATION KINETICS: PROPERTIES

- The rate of elimination is maximal and independent of drug concentration.
- Elimination is zero-order.
- **A constant amount of drug is eliminated per unit time.**
- Recall that in first-order kinetics a constant fraction of drug is eliminated per unit time.

SATURATION KINETICS: PROPERTIES

For a drug with zero-order elimination the plot of plasma concentration vs time is **linear.**

The drug is removed at a constant rate independent of plasma concentration.



SATURATION KINETICS: PROPERTIES

- Clearance is not constant.
- Clearance varies with the concentration of drug.

$$\text{Rate of elimination} = \frac{V_{max} \times C}{K_m + C}$$

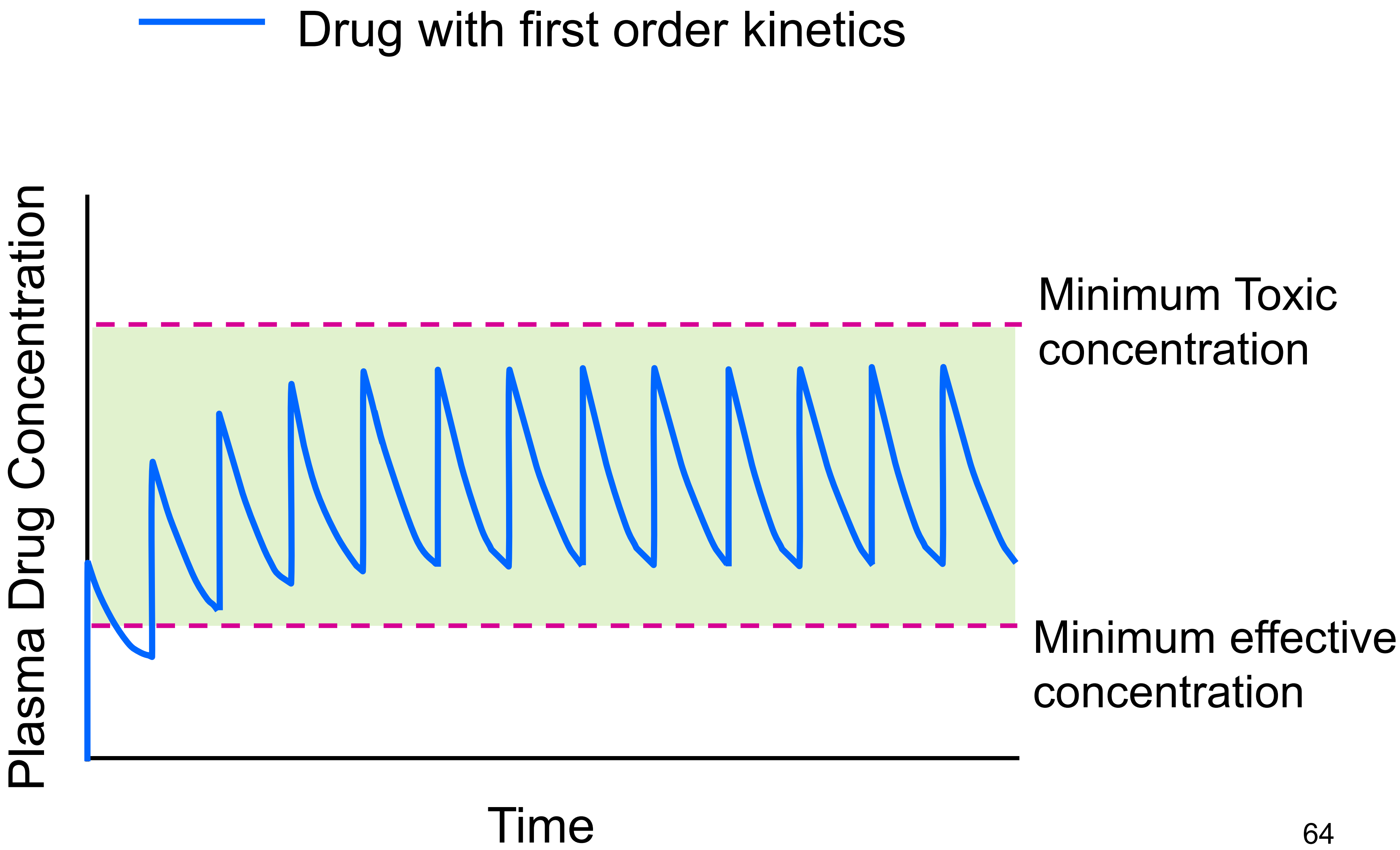
$$CL = \frac{\text{Rate of elimination}}{C} = \frac{V_{max}}{K_m + C}$$

SATURATION KINETICS: PROPERTIES

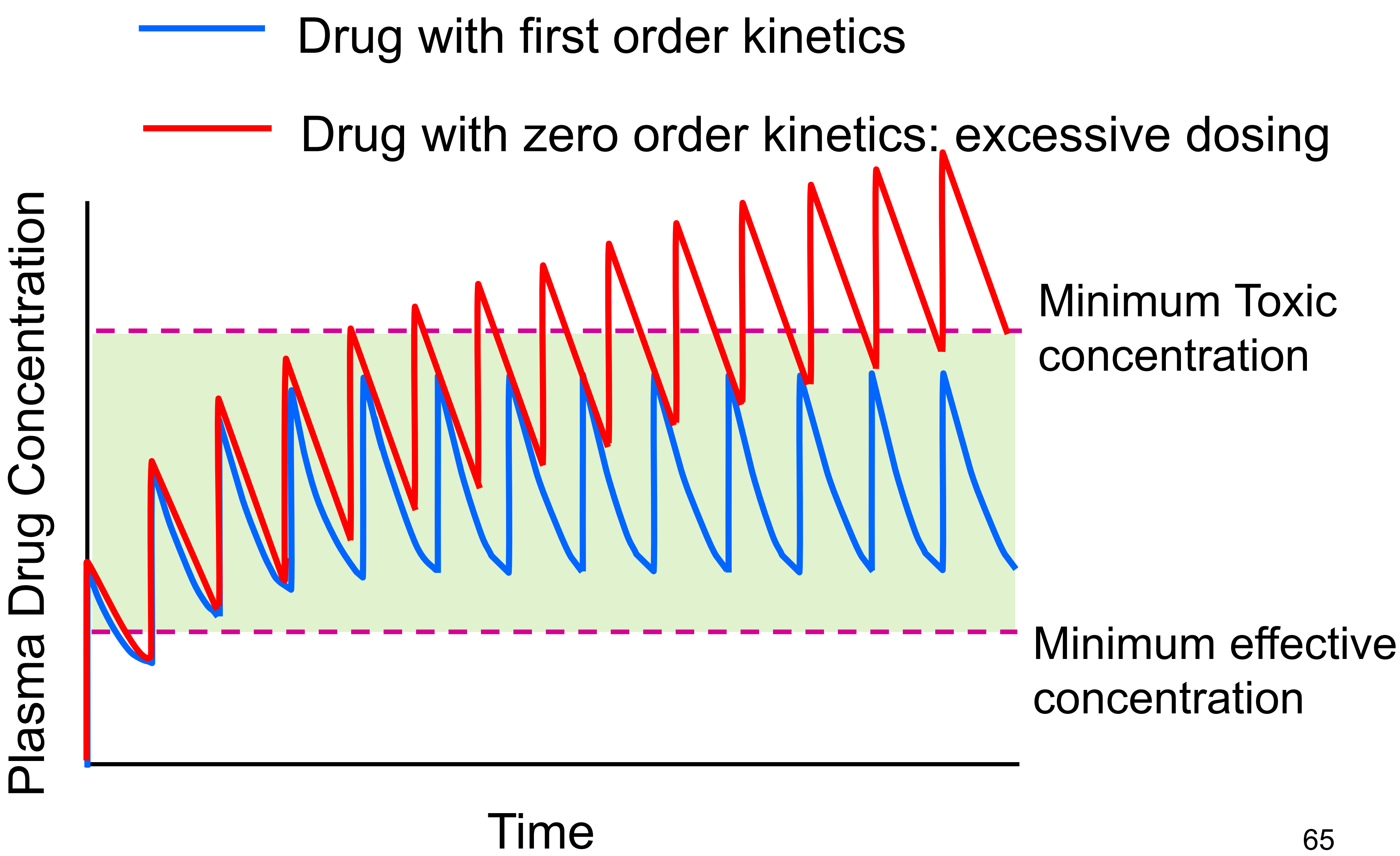
- Half-life is not constant.
- Therefore, the concept of "4 half-lives to steady state" is NOT applicable for drugs with non-linear elimination kinetics.

$$t_{1/2} = \frac{0.693 \times V_d}{CL}$$

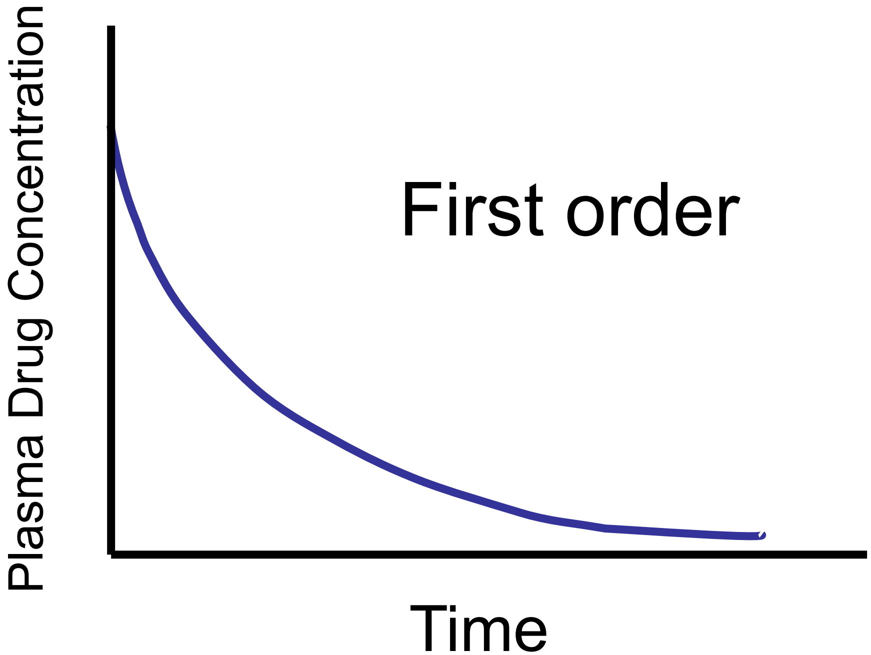
SATURATION KINETICS & DRUG TOXICITY

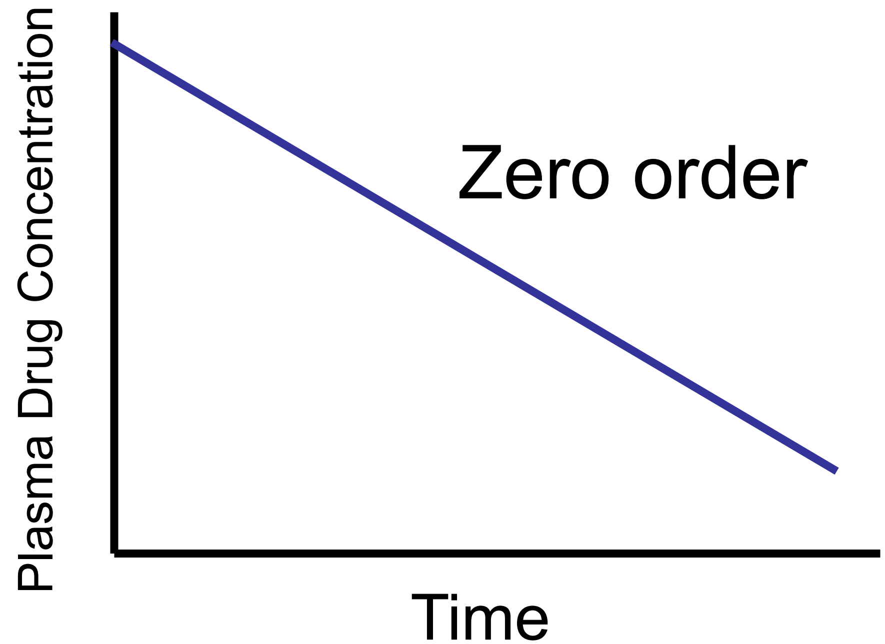


SATURATION KINETICS & DRUG TOXICITY



FIRST ORDER & ZERO ORDER KINETICS





- Constant fraction of drug eliminated per unit time
- Rate of drug elimination proportional to drug plasma concentration

- Constant amount of drug eliminated per unit time
- Rate of drug elimination independent of drug plasma concentration

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